



Fasting: A Complex, Double-Edged Blade in the Battle Against Doxorubicin-Induced Cardiotoxicity

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Abstract

In recent years, there has been a surge in the popularity of fasting as a method to enhance one's health and overall well-being. Fasting is a customary practice characterized by voluntary refraining from consuming food and beverages for a specified duration, ranging from a few hours to several days. The potential advantages of fasting, including enhanced insulin sensitivity, decreased inflammation, and better cellular repair mechanisms, have been well documented. However, the effects of fasting on cancer therapy have been the focus of recent scholarly investigations. Doxorubicin (Dox) is one of the most widely used chemotherapy medications for cancer treatment. Unfortunately, cardiotoxicity, which may lead to heart failure and other cardiovascular issues, has been linked to Dox usage. This study aims to comprehensively examine the possible advantages and disadvantages of fasting concerning Dox-induced cardiotoxicity. Researchers have investigated the potential benefits of fasting in lowering the risk of Dox-induced cardiac damage to solve this problem. Nevertheless, new studies indicate that prolonged alternate-day fasting may adversely affect the heart's capacity to manage the cardiotoxic properties of Dox. Though fasting may benefit overall health, it is essential to proceed cautiously and consider the potential risks in certain circumstances.

Keywords Doxorubicin · Cardiotoxicity · Fasting · Calorie restriction · Left ventricular mass

Abbreviations

Dox	Doxorubicin
TRF	Time-restricted feeding
IF	Intermittent fasting
FMD	Fasting-mimicking diet
ADF	Alternate day fasting
CVD	Cardiovascular disease
HFCD	High-calorie diet

CLPs	Common lymphoid progenitor cells
TILs	Cytotoxic CD8 ⁺ tumor-infiltrating lymphocytes
TFEB	Transcription factor EB
GDF15	Growth differentiation factor 15
T2DM	Type 2 diabetes mellitus

Doxorubicin (Dox) is a widely used chemotherapeutic agent employed on a worldwide scale for the treatment of various malignancies. The malignancies included in this group

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comprise, but not exclusively, cervical cancer, breast cancer, osteosarcoma, acute leukemia, and esophageal cancer [1]. Dox is widely recognized for its potent anti-cancer activity, primarily carried out through the intercalation of DNA and the inhibition of the topoisomerase II enzyme in fast-proliferating tumors [2]. These actions effectively halt the spread and growth of cancerous cells. Nevertheless, Dox has been associated with an adverse effect known as cardiotoxicity, which involves the potential harm inflicted on the myocardium. This cardiotoxicity has been linked to an increased risk of mortality among cancer patients, thereby posing restraints on its broad clinical application [3]. Studies have demonstrated that approximately 10–25% of patients administered with high doses of Dox develop clinically significant cardiac dysfunction. The mortality rate attributed to cardiotoxicity typically ranges from 1–5% [4]. Several mechanisms have been proposed to elucidate the process of Dox-induced cardiotoxicity. Among the most widely reported are oxidative stress, the generation of free radicals, and apoptosis [5]. While different, these mechanisms contribute to the overall damaging effect on the heart. Beyond these, additional mechanisms involved in Dox-induced cardiotoxicity have been identified. These include impaired mitochondrial function, perturbations in iron regulatory proteins, disruption of Ca^{2+} homeostasis, and autophagy [6]. Further, there are reports of the release of nitric oxide and inflammatory mediators, as well as altered gene and protein expression involved in apoptosis [7]. These changes, while complex, collectively contribute to the cardiotoxic effects observed with the use of Dox. Several factors, including age, gender, and underlying heart conditions, influence this sensitivity.

According to research data, a noticeable variation in patients' sensitivity towards Dox-induced cardiotoxicity occurs. Several factors, including age, gender, and underlying heart conditions influence this sensitivity [8, 9]. Further investigation into these factors has revealed that younger patients, particularly those under the age of 50, tend to exhibit a lower incidence of Dox-induced cardiotoxicity compared to their older counterparts. This trend is believed to be associated with the greater regenerative capacity of their myocardial cells [9, 10]. However, when younger patients do experience cardiotoxic effects, the impact can be more severe and potentially irreversible, necessitating vigilant monitoring and possibly the use of cardioprotective agents. Gender differences also play a significant role, with some studies indicating that females may be at a higher risk of developing cardiotoxicity from Dox than males [11]. The exact mechanism behind this disparity is not fully understood, but it is hypothesized that hormonal influences and differences in body composition, particularly in fat distribution and lean body mass, may contribute to the variance in sensitivity [12]. Underlying heart conditions such as pre-existing coronary artery disease, hypertension or a history

of heart failure significantly increase the risk of cardiotoxicity. These conditions can exacerbate the stress that Dox places on the heart leading to further damage. Consequently, patients with such conditions often require more careful dosing and closer cardiac surveillance during Dox therapy [13, 14]. Generally, it has been observed that certain groups are more susceptible to the cardiotoxic effects of Dox. These include patients who are older than 60 years of age, those diagnosed with hypertension, individuals who have diabetes, and patients with a pre-existing history of cardiac disease [9]. Thus, it is imperative for healthcare providers to consider these factors when prescribing Dox and to tailor treatment plans accordingly [15].

Three forms of Dox-induced cardiotoxicity may be identified based on their temporal pattern: acute, chronic, and delayed [16]. Acute cardiotoxicity is a common occurrence that usually manifests within a few days of Dox administration. The state is typically reversible and is distinguished by pericarditis, arrhythmias, and left ventricle dysfunction [17]. The frequency of chronic cardiotoxicity is comparatively lower than acute cardiotoxicity, which becomes apparent within a year. Chronic cardiotoxicity is linked with irreversible myocardial damage and an unfavorable prognosis [18]. It is significant to note that the manifestation of toxicity after chemotherapy may be delayed for several months or even years following the discontinuation of treatment [19, 20]. Despite the substantial amount of research and significant advancements in comprehending the molecular and pathophysiological mechanisms that give rise to cardiovascular toxicity due to cancer treatment, the exact mechanism accountable for the cardiotoxicity induced by Dox remains inadequately comprehended [21, 22].

Fasting Feasible in Cardiac Function

Fasting has garnered considerable attention in cardiology for a significant period [23]. A meta-analysis of the impacts of fasting on cardiovascular disease risk factors. Twelve studies were chosen for analysis via a systematic search of five databases. The study discovered that fasting holds the potential to enhance cardiometabolic function as a therapeutic approach to decrease overall body weight and target adipose tissue [24]. These effects of fasting might be due to several factors: energy metabolism and insulin regulation, autophagy, inflammation, and oxidative stress [25]. The heart relies on fatty acid oxidation for energy, as the body's glucose reserves are exhausted during fasting [26]. Along with this, fasting can cause insulin levels to drop [27]. Autophagy, a cellular repair mechanism, can also be stimulated in cardiomyocytes via fasting [28]. Furthermore, fasting has been shown to activate the cell's antioxidant defense system, which in turn helps prevent cardiac injury

by decreasing inflammatory factor levels and inflammation. In addition to these protective mechanisms, fasting has been observed to enhance mitochondrial function within the heart muscle cells [29, 30]. Another intriguing aspect of fasting is its potential to modulate the gut microbiome, which has been increasingly recognized for its impact on systemic health, including cardiovascular function. It is also worth noting that fasting can influence the autonomic nervous system, shifting the balance towards increased parasympathetic activity. This shift can lead to a reduction in heart rate and blood pressure, both of which are beneficial for cardiovascular health [31, 32]. Notably, the precise effects of fasting on the cardiovascular system remain debatable and may differ among individuals.

At present, this practice can be categorized into three distinct types that are commonly employed, namely time-restricted feeding (TRF), intermittent fasting (IF), and fasting-mimicking diet (FMD) [33]. TRF is a dietary pattern that limits food intake to specific time windows during the day, typically lasting between 4 and 12 h, without imposing caloric restrictions [34]. A study has shown that short-term TRF did not result in higher cardiovascular performance in this group of healthy individuals. However, minor improvements were found in functional capacity (endurance) and glucose tolerance [35].

IF, known as periods of voluntary abstention from food and drink, is characterized by recurrent, one- to several-day bouts of low or non-existent calorie intake [36]. Although several IF regimens have been suggested, alternate day fasting (ADF) and two days of fasting or severe restriction followed by five consecutive days of ad libitum eating are the two most well-known and researched techniques [37]. A clinical investigation examined the impact of IF on right ventricular function in individuals with pulmonary arterial hypertension. The findings suggest that IF directly impacts the function of the right ventricle and causes modifications to the composition of the gut microbiota [38]. FMD is a type of fasting that involves consuming a low-calorie, low-protein, high-fat diet for a few days. This fasting method has gained attention recently for its potential health benefits in reducing inflammation, improving blood pressure and cholesterol levels, promoting weight loss, and boosting overall metabolic health [39]. Numerous scholarly investigations have presented empirical data supporting the efficacy of fasting in the preservation of heart function and the reduction of cardiovascular disease (CVD) risk [40–42]. TRF or IF has been reported to mitigate the age-related deterioration in heart function, improve systolic blood pressure, and maintain cardiac structure in *drosophila* and mice [43, 44]. Implementing an FMD has been shown to mitigate the impact of a high-calorie diet (HFCD) on cardiometabolic risk and lifespan [45]. To summarize, FMD cycles have enhanced

cardiac vascularity and function and improved cardiotoxin resistance. Additionally, these diet cycles have shown the ability to avoid hyperglycemia, hypercholesterolemia, and hyperleptinemia, which depend on an HFCD. Further work is necessary to clarify the underlying processes contributing to the favorable benefits of FMD in counteracting the adverse consequences of a high-fat, high-calorie diet. Consistent with the findings, 2-day IF is an efficient approach for avoiding cardiometabolic illnesses in people with metabolic syndrome, requiring only moderate calorie restriction and minimal adverse effects. While compared to daily calorie restriction, randomized clinical research demonstrated that ADF did not improve adherence, weight loss, weight maintenance, or cardioprotection [46, 47]. Consequently, this physiological response may impede the body's ability to combat infections and heighten its susceptibility to cardiovascular ailments.

The efficacy of fasting elicits varying degrees of response [48]. For example, children require a different nutritional approach compared to adults due to their rapid growth and development. Their bodies are still building tissues and organs, and they have higher energy and nutrient needs relative to their size. Therefore, fasting might not be as beneficial for them and could potentially interfere with their growth and development if not carefully managed [49, 50]. On the other hand, the elderly often experience a decline in metabolic processes and may have chronic health conditions that fasting could exacerbate. However, some studies suggest that intermittent fasting could have positive effects on longevity and age-related diseases in this population. It is crucial to tailor fasting protocols to the individual's health status, ensuring they receive adequate nutrition to support their overall well-being [51, 52]. Those with pre-existing cardiovascular conditions present a particularly complex scenario. Fasting can have varying impacts depending on the specific condition and its severity. For example, individuals with stable angina might benefit from the anti-inflammatory effects of fasting, which could potentially reduce the risk of future cardiac events [53]. However, those with severe heart conditions or who are taking medications that require food intake for proper absorption might find fasting challenging or even dangerous. In all these patient subgroups, it is essential to consider the psychological impact of fasting as well [40]. Children may find it difficult to adhere to fasting regimens, potentially leading to stressor anxiety. The elderly might feel isolated if their social eating habits are disrupted, and individuals with cardiovascular conditions could experience increased anxiety about their health during fasting periods [54]. Ultimately, any fasting regimen should be personalized and closely monitored by healthcare professionals who can adjust the approach based on the patient's response and overall health.

Fasting's Potential in Cancer Prevention

Nutritional factors, such as glucose and amino acids, have a significant role in driving a variety of hallmark cancer cell characteristics, including fast proliferation, inactivation of tumor suppressor genes, invasion of neighboring tissues, and distant metastasis [55]. Fasting's role in cancer prevention is multifaceted, with effects primarily seen in regulating signaling pathways and molecular markers [56]. One of the fundamental mechanisms through which fasting can potentially inhibit tumor growth is by influencing the Insulin-like Growth Factor 1 (IGF-1) and mammalian Target of Rapamycin (mTOR) signaling pathways [57]. These pathways, when disrupted, have been associated with a decrease in the growth and proliferation of cancer cells. Fasting simultaneously increases oxidative stress levels, decreases circulating tumor DNA levels, and further lowers tumor risk. Nevertheless, fasting may not be appropriate for every cancer patient, and it is essential to create a tailored dietary regimen based on individual circumstances [58]. The safety and feasibility of using fasting in combination with standard cancer treatments are studied through clinical trials. As a result, the theoretical foundation for fasting-based therapy is the rapidly expanding body of research on the metabolic features of cancer since the turn of the 21st century [59]. Activating stress resistance and repair pathways, eliminating damaged and malfunctioning cells, and other processes may explain why fasting has been demonstrated to lower cancer incidence in clinical and pre-clinical studies [48]. It is plausible that specific markers, such as insulin and blood glucose, have declined, while various intracellular signaling pathways implicated in the onset and progression of cancer have been suppressed [60].

A clinical experiment was conducted to investigate the potential anticancer properties of IF from dawn to sunset in individuals with metabolic syndrome. This research included 14 volunteers diagnosed with metabolic syndrome who adhered to a fasting regimen and abstained from all food and beverages for 14 h a day for four weeks. The results of this study indicate that implementing IF throughout the period from sunrise to sunset has a direct impact on the regulation of specific genes. Consequently, it may be a supplementary therapeutic approach for individuals with metabolic syndrome [61]. By influencing systemic metabolism and bolstering antitumor immunity, cyclic fasting, or FMD, improves the efficacy of antineoplastic therapy in tumor-bearing animals [62]. Therefore, a clinical experiment was launched to examine the safety and biological impacts of a cyclic, five-day FMD when used with traditional anticancer therapy. The results show that cyclic FMD is well tolerated and causes significant

systemic metabolic changes in patients. Furthermore, FMD can modify systemic and intratumor immunity, ultimately triggering several immune responses that are effective against tumor growth. Phase II/III clinical studies are necessary to examine the anticancer activity and effectiveness of FMD [63].

Supplementing Cancer Chemotherapy with Fasting

Extensive preclinical research has shown the potential of fasting to confer protection onto healthy cells against many stresses, including chemotherapy. This protective effect is accomplished by concurrently increasing the susceptibility of cancer cells to chemotherapy and other therapeutic interventions [64]. During fasting, the body's viable cells can prioritize their efforts toward healing and self-repair. In contrast, malignant cells cannot enter this protective state due to oncoprotein activity, rendering them unable to acclimate to circumstances of limited nutrients [48, 65]. Fasting has been seen to result in the deprivation of nutrients, growth hormones, and other essential elements for proliferating cancer cells. Consequently, this deprivation leaves cancer cells more susceptible to cancer treatment, leading to an enhanced rate of cell death [66]. Preliminary clinical investigations have shown that incorporating fasting into chemotherapy regimens is safe and well-tolerated, with potential benefits in mitigating treatment-related toxicity [67, 68]. Firstly, fasting is known to induce autophagy, a cellular process responsible for degrading and recycling damaged cellular components [69]. This process can help clear out dysfunctional mitochondria and other cellular debris that accumulate because of Dox exposure, potentially reducing the overall cardiotoxic effect [70]. Secondly, fasting has been shown to modulate various signaling pathways implicated in cardioprotection. For instance, fasting can activate the sirtuin (SIRT) family of proteins, particularly SIRT3, which is localized in the mitochondria and plays a crucial role in maintaining mitochondrial integrity and function. Enhanced SIRT3 activity has been associated with reduced oxidative stress and improved cardiac function, both adversely affected by Dox [71, 72]. Moreover, fasting can influence the metabolism of Dox itself. It has been suggested that fasting may alter the pharmacokinetics of the drug, potentially reducing its concentration in heart tissue or altering its distribution within the body. This could lead to a decrease in direct toxic effects on cardiomyocytes caused by Dox [72, 73].

A multicenter, open-label, randomized DIRECT study provides further evidence supporting the potential of FMD as a viable and beneficial complement to chemotherapy in the treatment of women diagnosed with early-stage breast cancer [74]. The research outcomes demonstrate that a 48-hour fasting period is well-tolerated and does not lead to increased weight loss, hospital

admissions, or reductions/delays in chemotherapy dosage. The implementation of fasting was associated with reduced treatment adjustments and improved quality-of-life ratings throughout the treatment period. A comprehensive research study has provided intriguing evidence that the practice of intermittent fasting when coinciding with the adjuvant chemotherapy AC protocol, which includes the use of medications like Dox and cyclophosphamide, could potentially offer differential stress resistance to patients who have breast cancer [75]. Additionally, a clinical study suggests that FMD may enhance chemical immunotherapy or chemotherapy for patients with advanced Triple-Negative Breast Cancer (aTNBC) or early TNBC [76].

Simultaneously, the efficacy of chemotherapy on the tumor may be enhanced by fasting in mice via cellular immunity [77, 78]. According to research, the combination of chemotherapy and an FMD enhances the levels of bone marrow common lymphoid progenitor cells (CLPs) and cytotoxic CD8⁺ tumor-infiltrating lymphocytes (TILs), resulting in a significant delay in the progression of breast cancer and melanoma [79]. A recent investigation further showed that the fasting-mimicking diet synergizes with ferroptosis against quiescent cells resistant to chemotherapy [80].

Genetic predispositions that influence susceptibility to Dox-induced cardiotoxicity include polymorphisms in genes involved in drug metabolism, such as those encoding for cytochrome P450 enzymes, and genes involved in cardiac function, such as those encoding for ion channels and contractile proteins [81]. Variants in these genes can alter the pharmacokinetics of Dox or the susceptibility of cardiomyocytes to damage, thereby increasing the risk of cardiotoxicity. Epigenetic mechanisms, such as DNA methylation, histone modification, and non-coding RNA expression, can also modulate gene expression without altering the underlying DNA sequence [82, 83]. These mechanisms are sensitive to environmental changes and cellular stress and can be influenced by fasting. For example, fasting can lead to changes in histone acetylation patterns which affect the expression of genes involved in stress response and cell survival. By altering these epigenetic marks, fasting may potentially reduce the expression of genes that promote cardiotoxicity and increase the expression of protective genes [84]. Further research is needed to fully understand the molecular mechanisms involved and to develop fasting protocols that can be safely and effectively integrated into cancer treatment regimens. Tailoring interventions based on an individual's genetic and epigenetic profile could pave the way for personalized approaches to chemotherapy, optimizing efficacy while minimizing adverse effects.

Risks and Benefits of Fasting During Chemotherapy

Fasting is primarily driven by the potential benefits this practice could provide patients undergoing the often-grueling chemotherapy process [85]. These benefits may include, but are not limited to, improved tolerance to treatment, enhanced effectiveness of the drugs, and reduced side effects. However, this method has drawbacks, presenting a significant challenge in striking the right balance between potential health benefits and safety [37, 86]. The complexity of this issue lies in the delicate balance that needs to be maintained during treatment and the potential risks that come with depriving the body of nutrients during such a critical time. A randomized controlled study assessed the safety and acceptability of water-only fasting among women with gynecologic malignancies undergoing a minimum of six scheduled rounds of chemotherapy [87]. Nevertheless, it is crucial to acknowledge that fasting may not suit every individual diagnosed with cancer. The act of fasting has the potential to pose risks for those who are already susceptible to malnutrition or dehydration, and it has the potential to impede the absorption and efficacy of some drugs. Therefore, it is essential to conduct more studies to comprehensively comprehend the underlying processes of this occurrence and ascertain the most effective use of fasting in cancer chemotherapy treatment (Table 1 [75, 88–92]).

The interaction between fasting and Dox-induced cardiotoxicity is a complex phenomenon that involves intricate cellular and molecular pathways. One critical aspect of this relationship is the modulation of the immune system, encompassing both inflammatory responses and immune regulatory mechanisms. Fasting has been shown to induce several physiological changes that can influence the immune system. During fasting periods, the body shifts from glucose metabolism to fatty acid oxidation and ketone body production for energy [93, 94]. This metabolic switch has been associated with a reduction in systemic inflammation. For instance, fasting can decrease the levels of pro-inflammatory cytokines such as tumor necrosis factor-alpha (TNF- α) and interleukin-6 (IL-6), both known to exacerbate tissue damage and impair cardiac function. By attenuating these inflammatory responses, fasting may confer a degree of protection against the cardiotoxic effects of Dox [95]. Moreover, fasting can influence immune modulation by promoting autophagy, which are particularly susceptible to Dox-induced damage. Enhancing autophagic activity through fasting may help mitigate the cardiotoxic effects of the drug by facilitating clearance of damaged mitochondria and reducing oxidative stress, a key mediator of Dox-induced cardiomyopathy [96]. However,

Table 1 Fasting in combination with Doxorubicin in Cancer Treatment

Doxorubicin treatment	Model	Disease occurrence	Information of fasting	Specific biomarker alteration	Function	References
Pharmacologic-grade doxorubicin hydrochloride	Mice	Pediatric cancer	24 h of fasting alternating with 24 h of feeding	Nuclear translocation of transcription factor EB (TFEB) and growth differentiation factor 15 (GDF15)	Their studies reveal that both sustained alternate-day fasting and the TFEB/GDF15 pathway aggravate doxorubicin cardiotoxicity.	[88]
Oral administration of doxycycline (DOX, 2 g/l in water)	Mice	Breast cancer, lung cancer and cervical cancer	Intermittent fasting for 48 h from day 16 and day 22, respectively	GSK3 β –SIRT7 axis	The conclusions of their research suggest that targeting the GSK3 β –SIRT7 axis could potentially mimic the effects of fasting on tumor growth and metastasis.	[89]
i.v. twice a week over a three-week period	Mice	Hepatocellular carcinoma (HCC)	Fast for 12 h every night, the short-time fasting (STF) group was fasted once.	VEGF receptor 2	These findings illustrate that the advantageous impact of fasting on the outcomes of anticancer therapy extends beyond the modulation of metabolism at the molecular level.	[90]
Intraperitoneal inoculation of 15 mg/kg of doxorubicin	Mice	Intestinal, stomach and pancreatic tumors	Fasted for 24 h before and 24 h after administration of doxorubicin	Blood levels of glucose and the ketone body β -hydroxybutyrate, as well as their body weight	The findings demonstrate a fasting-associated mechanism for lipid homeostasis and offer potential biomarkers for predicting the protective effects of fasting against chemotherapy toxicity.	[91]
Doxil at the dose of 10 mg/kg	Mice	Breast cancer and ovarian cancer	Low-calorie and low-protein named fasting-mimicking diet (FMD)	Cell cycle analysis and in vivo antitumor analysis	These findings provide evidence that the FMD strategy holds significant potential as a practical and promising adjunct to single nanodrug therapy for enhancing cancer treatment and facilitating clinical translation.	[92]
Chemotherapy AC (doxorubicin, cyclophosphamide) protocol	Human	Breast cancer	The participant will fast intermittently for three consecutive days around chemotherapy, fasting for 18 h a day from 12 am to 6 pm. They will eat during a 6-hour window from 6 pm to 12 am, with permission to drink water during the fasting hours.	Metabolic parameters, hematologic parameters and inflammatory marker	Implementing fasting during chemotherapy was well tolerated and resulted in a decrease in chemotherapy-related toxicity.	[75]

it is important to note that the relationship between fasting, the immune system, and Dox-induced cardiotoxicity is complex and not entirely understood. While fasting may offer some protective benefits, it is also possible that it could impair the immune system's ability to respond to infections or other stressors, particularly in cancer patients who are already immunocompromised [97]. Furthermore, the timing, duration, and frequency of fasting, as well as the individual patient's health status, can significantly influence the outcomes. In addition to autophagy, fasting can also modulate the immune system by altering the composition and function of immune cells. For example, it can lead to a reduction in circulating lymphocytes and a shift towards a more anti-inflammatory immune profile. Such change may be beneficial in mitigating Dox-induced cardiotoxicity by potentially decreasing the likelihood of an immune-mediated inflammatory response that could further harm heart tissues [28, 98].

Overall, there exists a complex and not fully comprehended interplay between fasting patterns with regards to how they affect immunological responses. Despite potential protective effects of fasting against Dox-induced cardiotoxicity; conversely an impaired immune response towards infections or other challenges may be observed especially among already compromised cancer patients [29, 99]. It should be considered that various factors such as timing/duration/frequency of fasts and individual health conditions significantly impact on eventual outcomes (Fig. 1).

Can Fasting Minimize Dox Cardiotoxicity?

Fasting may improve the body's capacity to metabolize and tolerate Dox, minimizing its harmful effects on the heart. The production of β -hydroxybutyrate (BHB) is particularly significant when fasting, as the body switches from glucose to fat as its primary energy source [41]. It has been

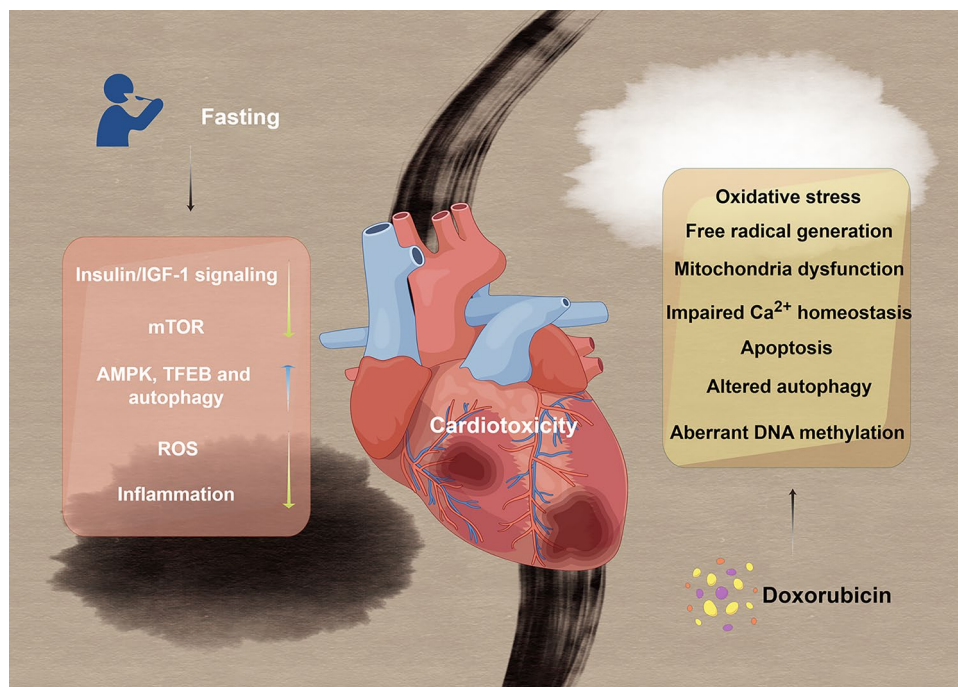


Fig. 1 The potential molecular effects of fasting in Dox-induced cardiotoxicity. Fasting, a state of voluntary abstinence from food and drink, has been shown to have several effects on the body's cellular processes. One notable effect is the downregulation of mTOR (mammalian target of rapamycin) and IGF1 (Insulin-like Growth Factor 1) signaling, two critical pathways involved in cell growth and metabolism. Further, fasting has been shown to induce AMPK (AMP-activated protein kinase) and TFEB (Transcription Factor EB), which upregulates the expression of autophagy genes in the heart. This suggests a potential protective mechanism of fasting on heart health. In addition to these effects, fasting has also been linked to decreased inflammation, a critical factor in many chronic diseases. On the other side of the spectrum, Doxorubicin (Dox), a chemotherapy drug, has

been associated with cardiotoxicity, a harmful effect on the heart's muscle tissue, known as the myocardium. This adverse effect has profound implications, as it has been linked to an increased risk of mortality among cancer patients. This raises significant concerns and limitations regarding its widespread clinical application. Several mechanisms have been proposed to understand the process of Dox-induced cardiotoxicity. Among the most widely reported are oxidative stress, which involves an imbalance of antioxidants and free radicals in the body, the generation of harmful free radicals, and apoptosis, the process of programmed cell death. These mechanisms provide insight into the potential pathways through which Dox causes harm to the heart. All Figures were created with FigDraw

confirmed to play a cardioprotective role, particularly in conditions of ischemia and hypertension. According to scientific research, BHB might provide a defense against Dox-induced cardiotoxicity [100]. The proposed protective role of BHB against DOX-induced cardiotoxicity is thought to be achieved through inhibiting cell apoptosis and preventing oxidative stress. Moreover, BHB is also believed to play a crucial role in preserving the integrity of the mitochondrial membrane.

On the flip side, fasting could exacerbate Dox-induced cardiotoxicity. Patients undergoing chemotherapy often experience significant weight loss and nutritional deficiencies [101, 102]. Adding fasting to this equation could further compromise their nutritional status and overall health, potentially making them more susceptible to the cardiotoxic effects of Dox. Moreover, the impact of fasting on Dox-induced cardiotoxicity may vary depending on several factors, including the patient's overall health, the type and stage of cancer, and the specific chemotherapy regimen.

Autophagic dysfunction resulting from lysosomal damage is a mechanistic hallmark of Dox-induced cardiotoxicity [103]. Autophagy is a process that allows cells to recycle their own components, including proteins and organelles. When autophagy is disrupted, damaged proteins and organelles accumulate, which can be toxic to cells. Lysosomes are organelles responsible for digesting and breaking down cellular waste, including damaged proteins and organelles [104]. Autophagic dysfunction and the buildup of toxic waste result from lysosome damage since the organelle can no longer perform its functions. Dox has been found to cause cardiotoxicity by damaging lysosomes in cardiac cells, which causes autophagic dysfunction and the buildup of toxic waste in the heart. Cardiotoxicity, or damage to the heart muscle, may result from this [105, 106]. Current scholarly investigations have been chiefly directed toward exploring the prospective therapeutic function of the transcription factor EB (TFEB) in Dox-induced cardiotoxicity. TFEB serves as a critical regulator of the autophagy-lysosomal pathway, a crucial cellular process involved in destroying cellular components that are either damaged or no longer required [107]. Previous research has demonstrated that ADF can rescue advanced forms of cardiomyopathy in mice by stimulating TFEB, a master regulator of autophagy and lysosomal biogenesis. Several research studies have also proven that fasting can reduce myocardial injury through mechanisms involving the autophagy-lysosome pathway [29, 108]. It has also been proposed that the overexpression of TFEB may improve the adverse effects of Dox on the heart [19, 109]. According to this paradigm, the potential of ADF to mitigate anthracycline-induced cardiotoxicity may be attributed to its ability to reinstate autophagy and enhance TFEB expression. Can the combination of ADF and Dox potentially mitigate the cardiotoxic adverse effects

of Dox during treatment? The research conducted by Ozcan M et al. revealed that the implementation of ADF during Dox therapy resulted in an upregulation of TFEB expression [88]. So, it was identified that ADF could exacerbate Dox cardiotoxicity. This research serves as a reminder for cancer patients treated with Dox that ADF may be detrimental to the efficacy of the treatment.

To examine the validity of this idea, the study team used adeno-associated virus-mediated gene transfer to induce overexpression of TFEB in cardiac tissue. The study's findings confirmed that excessive TFEB increased mortality in mice and reduced left ventricular mass in both standard and pathological scenarios. On the other hand, the targeted elimination of TFEB in mice has successfully mitigated Dox-induced cardiac dysfunction and myocardial atrophy. Prior research has demonstrated that TFEB can stimulate the growth differentiation factor 15 (GDF15) expression in macrophages through a specific mechanism [110]. GDF15 in plasma has been identified as a potential biomarker for the diagnosis of Dox cardiomyopathy [111]. By examining the plasma of mice used in the experiment, they further discovered that while ADF had a diminishing effect on the levels of GDF15, the administration of Dox resulted in a significant increase in GDF15 production. Consequently, the concurrent use of ADF and Dox resulted in higher levels of GDF15 than those observed with Dox treatment alone. Likewise, a significant increase in GDF15 synthesis was detected in the bloodstream of mice overexpressing TFEB. To ascertain the potential of GDF15 to cause myocardial atrophy, they employed exogenously supplementing recombinant GDF15 for verification purposes. The study's findings indicated that the administration of GDF15 did not significantly impact body weight. However, it was observed that GDF15 did lead to a substantial reduction in the left ventricular mass of the mice.

This study originated from clinical concerns and was subsequently validated through experimentation with the hypothesis that excessive activation of ADF or TFEB may result in the gradual decline of left ventricular mass via the GDF15 or MuRF1 pathway. This article highlights the necessity of controlling cachexia and left ventricular mass after Dox treatment and recommends heightened clinical vigilance regarding the correlation between calorie intake and cardiotoxicity from chemotherapy.

Conclusion

Recent studies have also indicated that fasting may have potential health risks that should be considered [112]. Empirical investigations have demonstrated that extended periods of fasting can elicit a stress response in the brain that may harm immune cells, rendering them less effective

in combating infections and elevating the likelihood of developing cardiovascular disease. Among overweight people with type 2 diabetes mellitus (T2DM), recent research found that a 12-week term of TRF resulted in a drop in LDL-C compared to HDL-C [113]. In males with prediabetes, LDL-C and HDL-C do not change after five weeks of TRF [114]. Therefore, it is crucial to customize clinical procedures based on the patient's clinical indication. Therefore, implementing extended periods of complete fasting is a significant problem for the human population. The efficacy of the fasting paradigm may not be universally applicable, and its execution may not promote the body's ability to withstand pathogenic invasions. The present study focuses on the correlation between calorie intake and Dox-induced cardiotoxicity, highlighting the need for increased clinical consideration. Further randomized controlled studies are required to examine the effects of intermittent fasting on Dox medicines, specifically exploring parameters like timing, frequency, and other relevant variables. The potential use of fasting as a strategy to mitigate or delay Dox-induced cardiotoxicity might enhance the efficacy of Dox as an anti-tumor therapeutic agent.

Dox, a potent chemotherapy drug used to treat various types of cancer, can have detrimental effects on the heart, leading to the development of cardiotoxicity, a condition characterized by the weakening of the heart muscle [115, 116]. Over the past few years, the medical and scientific community has taken an increasing interest in the potential therapeutic benefits of fasting [117]. Fasting, a practice of voluntarily abstaining from food for a certain period, has garnered significant attention in the medical community for its potential benefits in reducing the risk of Dox-induced cardiotoxicity [100]. During fasting, the body undergoes a metabolic adaptation, shifting from glucose metabolism to utilizing ketone bodies produced from stored fats. This metabolic switch has been shown to improve cardiac function and reduce oxidative stress, which are implicated in Dox-induced cardiomyopathy [118]. However, a recent study indicates that fasting may worsen the cardiotoxic effects of Dox. This discovery is crucial since it can change how patient care and treatment regimens are approached, especially for chemotherapy recipients. It is essential to consider this factor while creating a thorough treatment strategy [88]. Hence, it is crucial to thoroughly evaluate fasting treatment's possible impact on decreasing Dox-induced cardiotoxicity risk and to conduct customized assessments for each patient.

Concurrent therapies, such as statins and Angiotensin-converting enzyme (ACE) inhibitors, play a pivotal role in the management of cardiotoxic effects associated with Dox, particularly when administered in a fasting state. The cardioprotective mechanisms of these agents are multifaceted and synergistic, offering a comprehensive approach to mitigate the adverse cardiac effects of

this potent chemotherapeutic agent [119, 120]. Statins, primarily known for their lipid-lowering properties, have also demonstrated additional benefits in cardiovascular health. They exert their cardioprotective effects by reducing inflammation and oxidative stress, both of which are implicated in the pathogenesis of Dox-induced cardiotoxicity. By inhibiting the production of mevalonate, statins decrease the synthesis of isoprenoid intermediates that are essential for post-translational modification of small GTP-binding proteins involved in cell signaling and survival [121]. This action can help preserve myocardial cell integrity and function, counteracting some of the cellular damage caused by Dox. ACE inhibitors, on the other hand, are well-established in treating hypertension and heart failure. Furthermore, ACE inhibitors have been demonstrated to possess cardioprotective properties beyond their hemodynamic effects. They can increase the bioavailability of bradykinin, which in turn stimulates the production of nitric oxide and prostaglandins, resulting in vasodilation and improved coronary blood flow [122, 123]. Additionally, ACE inhibitors can reduce myocardial fibrosis by decreasing the activity of the renin-angiotensin-aldosterone system (RAAS), which is known to contribute to the progression of heart failure. When administered concurrently with Dox, especially in a fasting state, these agents may provide enhanced protection against cardiotoxicity. Fasting has been proposed to enhance the efficacy of specific chemotherapeutic agents by altering pharmacokinetics and reducing gastrointestinal toxicity, potentially allowing for higher chemotherapy doses [124, 125]. However, fasting can also impact drug metabolism and disposition, necessitating careful monitoring and potential dose adjustments for concurrent medications such as statins and ACE inhibitors. In conclusion, simultaneous use of statins and ACE inhibitors with Dox holds promise as a strategy for reducing the risk of cardiotoxicity; particularly when applied within a fasting context [126]. Future studies will concentrate on establishing the optimal fasting schedule and length: Examine the optimal fasting cycle and length to provide the highest level of cardiac protection based on the patient's health and type of cancer [127].

In conclusion, fasting represents a fascinating area of research in the context of reducing the risk of Dox cardiomyopathy. While preliminary evidence suggests potential benefits, further studies are needed to clarify the optimal fasting protocols and determine the long-term effects on treatment outcomes. By gaining a deeper understanding of the mechanisms underlying the double-edged impact of fasting on cardiovascular, we can harness the benefits of fasting to optimize Dox treatment and improve patient outcomes.

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Declarations

Conflict of interest The authors have not disclosed any competing interests.

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